

Politecnico di Torino

Collegio di Elettronica, Telecomunicazioni e Fisica

# Integrated Systems Technology
# Report - Final Project
## SOI P-MOS

Master degree in Electronic Engineering

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# Contents


**1 Introduction** **1**

**2 Pen & paper design** **3**

**3 Fabrication of the device** **5**
3.1 Substrate definition . . . . . 6
3.2 Gate creation . . . . . 7
3.3 LDD . . . . . 9
3.4 Spacers . . . . . 12
3.5 HDD . . . . . 13
3.6 Aluminum contact . . . . . 14
3.7 Complete structure . . . . . 15

**4 Electrical Characterisation** **17**
4.1 Simulation . . . . . 17
4.2 FOM's value estimation . . . . . 18
4.2.1 $I_{ON}$ & $I_{OFF}$ . . . . . 18
4.2.2 SS . . . . . 19
4.2.3 $V_{th}$ . . . . . 19
4.2.4 DIBL . . . . . 21
4.2.5 Values . . . . . 21
4.3 Considerations on the obtained values . . . . . 21

**5 Conclusion** **23**

**A Planar pMOS** **24**

**Bibliography** **27**

# CHAPTER 1

# Introduction

The project revolved around the creation of a P-MOS on a SOI wafer with a nominal channel length of 80 nm. To achieve this goal we decided to start from designing a planar pMOS transistor, equivalent to the nMOS it was given us during the first laboratory experience, then move first to an SOI FD-pMOS and lastly to an UTTB SOI FD-pMOS to improve some parameters we did not consider acceptable.
The planar design is briefly presented in Appendix A, while in the report chapters we focused on explaining the design of the SOI. We decided to follow the history of our project, thus focusing on the standard SOI and in the end of the report move to the UTTB explaining the reasoning behind this step.

This report will be divided in different parts:

1. **Pen & paper design**: this part is focused on the design made by us before the actual creation of the P-MOS with the Sentaurus software.
2. **Fabrication of the device**: this is the chapter dedicated to the structure's creation by using Sentaurus. Inside this chapter will be discussed all the steps needed during the fabrication and their purpose.
3. **Electrical Characterisation**: this chapter is dedicated to the results of the simulation.
4. **Conclusion**: this chapter will be dedicated to the conclusions.

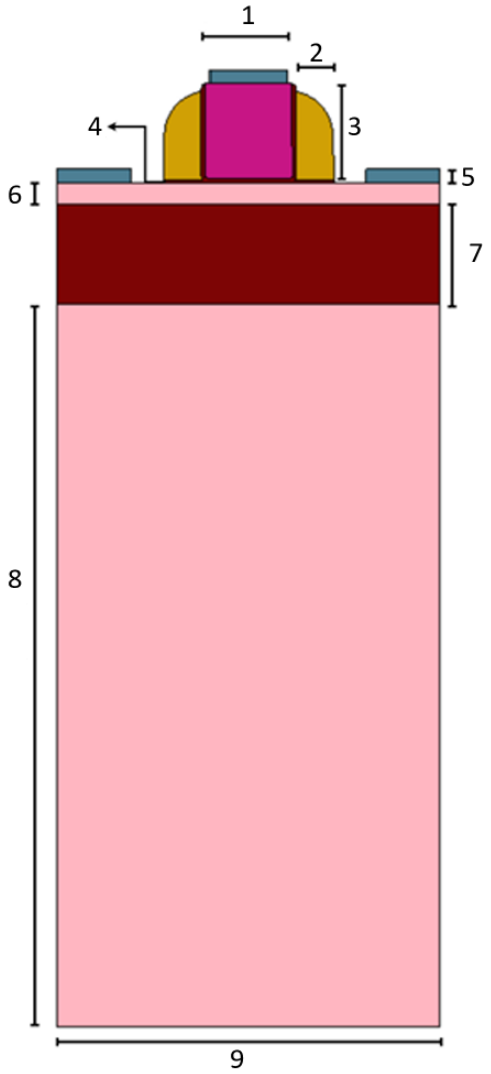
In the following part will be reported the dimensions of the wafer, the dimensions of the structure and also the technological parameters used during the fabrication. Tab 1.1 shows only the implants. All the other processes like RTA, deposition, etching, etc... are reported and discussed in Chapter 3.

| Technological Parameters | | | | | |
|---|---|---|---|---|---|
| **Parameter** | **Element** | **Dose ($cm^{-2}$)** | **Energy (keV)** | **Angle** | **Discussed in Section** |
| Substrate doping | B | $1 \cdot 10^{15}$ | - | - | 3.1 |
| n-well (1) | As | $1 \cdot 10^{13}$ | 10 | 0° | 3.2 |
| n-well (2) | As | $1 \cdot 10^{12}$ | 2 | 0° | 3.2 |
| LDD | B | $4 \cdot 10^{14}$ | 4 | 0° | 3.3 |
| HDD | $BF_2$ | $4.5 \cdot 10^{15}$ | 10 | 7° | 3.5 |

Table 1.1: UTTB pMOS FD-SOI parameters

| Structure Dimensions | | | |
|---|---|---|---|
| **Number** | **Object** | **Value** | **Unit of Measurement** |
| 1 | Channel length | 80 | nm |
| 2 | Spacers width | 30 | nm |
| 3 | Gate height | 80 | nm |
| 4 | Gate Oxide thickness | 3.5 | nm |
| 5 | Aluminum contact thickness | 30 | nm |
| 6 | Silicon thickness | 25 | nm |
| 7 | Buried Oxide thickness | 50 | nm |
| 8 | Substrate thickness | 750 | nm |
| 9 | Wafer width | 400 | nm |

Table 1.2: UTTB pMOS FD-SOI dimensioning



# CHAPTER 2

# Pen & paper design

This chapter contains the photo taken to the pen and paper design made before the actual fabrication of the device.

By looking at Fig 2.1 different things can be noticed:

- All the doping required to build the needed regions (LDD, HDD, n-well) are specified since we have estimated beforehand the required dose, energy and element required
- An estimation of the minimum Gate Oxide thickness was made in order to build a sufficiently thick Oxide since we didn't want that the electric field to break it
- The theoretical threshold voltage of the device was calculated

What is missing from the picture are all the RTA processes that were needed during the fabrication process. This is due to the fact that we didn't know how to estimate neither the temperature at which the RTA has to be performed nor the time required. During the fabrication different tests were made and the best one was kept.

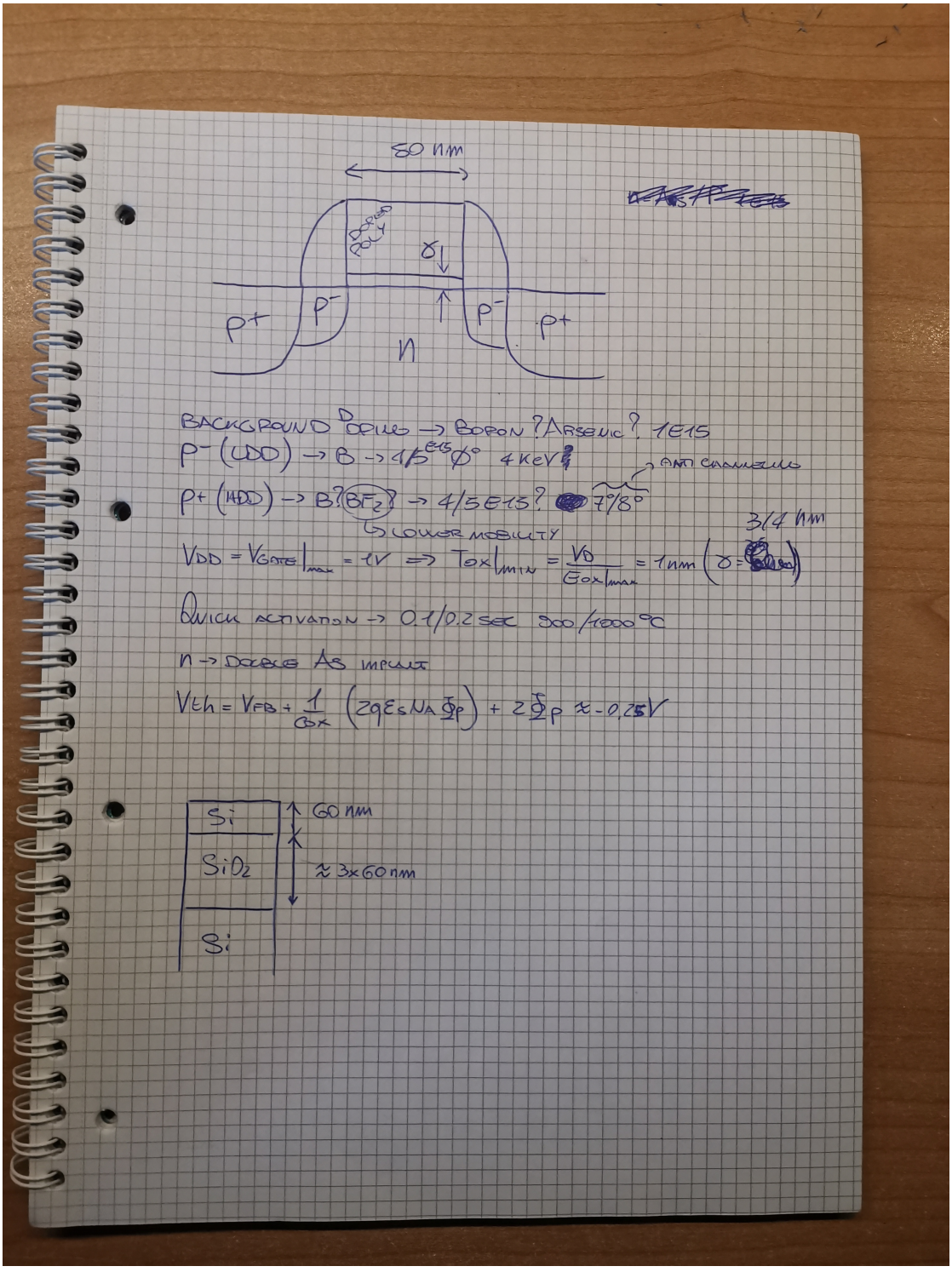



Figure 2.1: Pen & paper design

# CHAPTER 3

# Fabrication of the device

This chapter is dedicated to the description of the various steps needed to build the device. The reader is reminded of two things:

1 To reduce the required building time, only half of the structure was created for as many steps as possible. After the completion of the device the structure was mirrored in respect with the y axis to create the overall device.

2 All the analysis except for the last step are referred to the standard SOI. However, due to the reasoning explained in the end of the chapter, we concluded that no modifications were needed except for the dimensioning of the layers.

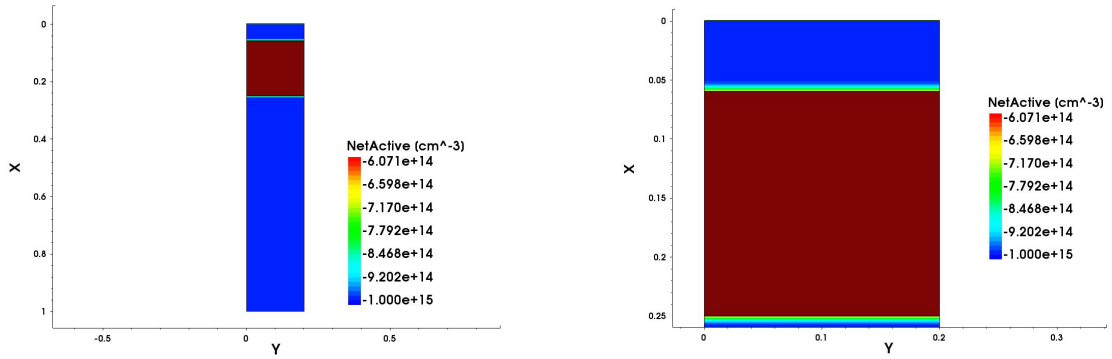
## 3.1 Substrate definition

To correctly simulate the P-MOS on SOI it was mandatory to correctly define the wafer that we had to work with.

The first step consisted in the definition of the size of the wafer and the definition of the substrate. We assumed that we had to work with a SOI wafer that had a p-type substrate doped uniformly with a concentration of $1 \cdot 10^{15}$ cm$^{-2}$ of Boron: this value has been chosen based on previous laboratory experience held during this course.

The thin Silicon film on top and the buried Oxide have a thickness of respectively 60 nm and 190 nm: these values has been chosen in order to directly progress from the planar device designed in the Appendix A to a SOI made with standard technology [1, 2]. Furthermore, these values have been confirmed by the professor during theoretical lessons to be suitable for the standard SOI typical dimensioning.

From Fig. 3.1a it easily noticeable the background doping, the buried Oxide layer and the this Silicon layer on the top part.




(a) (b)

Figure 3.1: Structure of the SOI wafer (a), enlarged version to highlight the top part of the wafer (b)

## 3.2 Gate creation

Once the substrate was correctly defined it became possible to start creating the first part of the P-MOS structure.

Firstly a double[1] Arsenic implant was made to dope the thin Silicon layer. Arsenic has been preferred to phosphorus due to its larger size and weight, which makes it a more suitable choice when it is required to obtain a sharp doping profile, as it diffuses less during high-temperature annealing. The Arsenic implant creates a sort of n-well which is required in order to properly create the p+ contact that will be implanted with the HDD process. After the implant a rapid activation (RTA - Rapid Thermal Annealing) was performed.

As a next step, the gate Oxide ($SiO_2$) was created. In order to grow a very thin and precise Oxide layer a dry oxidation was required.
Dry oxidation is preferred to wet oxidation when it is required to obtain a layer with better electrical quality, which is exactly the case of gate oxides. This difference is due to the out-diffusion of the H2 molecules, which create "voids" along the path of the wet oxides, thus making it less dense and therefore weaker. This is evident when comparing the two reactions:

$$Si + O_2 \rightarrow SiO_2$$
$$Si + 2H_2O \rightarrow SiO_2 + 2H_2$$

Thermal oxidation always requires high temperature (within the range of $700 - 1200$°C), in order to provide enough energy for the reaction to occur.
At 920°C, it can be seen from Fig. 3.2[2] that, for small time intervals, the growth rate of the oxide thickness is around 21nm/hour [3]. We therefore expected to obtain a 5nm oxide layer performing the oxidation at that temperature for about 15 minutes.

| Oxidation emperature (°C) | $A$ ($\mu$) | $B$ ($\mu^2$/h) | $B/A$ ($\mu$/h) | $\tau$ (h) |
|---|---|---|---|---|
| 1200° | 0.040 | 0.045 | 1.12 | 0.027 |
| 1100° | 0.090 | 0.027 | 0.30 | 0.076 |
| 1000° | 0.165 | 0.0117 | 0.071 | 0.37 |
| 920° | 0.235 | 0.0049 | 0.0208 | 1.40 |
| 800° | 0.370 | 0.0011 | 0.0030 | 9.0 |
| 700° | ... | ... | 0.00026 | 81.0 |

Figure 3.2: Rate constants for dry oxidation of Si

The final result was a bit off our expectations, therefore we used this as a starting point to be tuned until we reached our goal. This was achieved at a temperature of 850°C and a time

[1] Two different implants were performed at different energies to maintain better control of $V_{th}$ similarly to what is usually done for planar devices. We assumed two steps were enough, since punch-through is already controlled by the buried oxide.

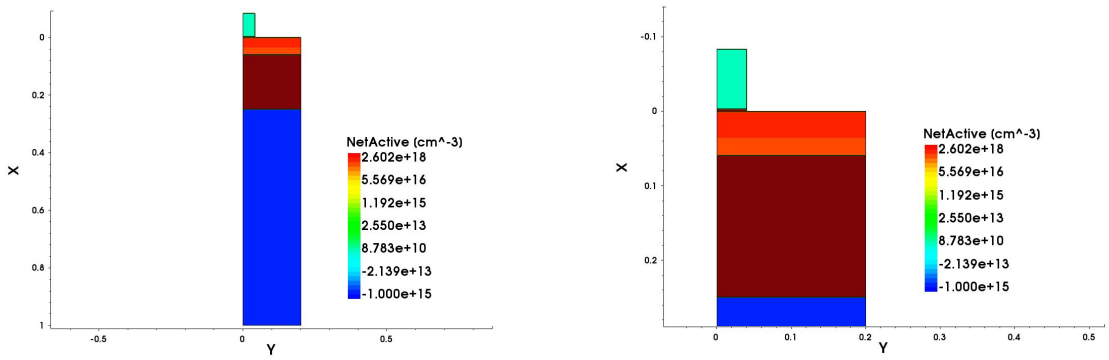
[2] Figure taken from "physics of technological process" course, prof. Matteo Cocuzza

[3] Deal-Grove model: at short time, the growth rate is limited by the rate of reaction of oxidant with silicon, which is mathematically described as $\frac{B}{A}t$

interval of 10 minutes. The corrections are probably due to the linear growth approximation and the neglect of the so-called "native oxide" layer.

Lastly, the Polysilicon (Poly) required for the Gate was deposited. By looking at Fig 3.3b it can be noticed that the Gate has a width of only 40 nm even if the requirement was a channel length of 80 nm. As previously mentioned, this is due to the fact that the structure will be mirrored in the last step. After the Poly deposition the first photolithographic process was performed in order to etch away the unwanted part of the Poly. Since Polysilicon can be used as a mask during the Oxide etching another etching step was performed to remove the thin Oxide layer previously created. Fig 3.3b shows that the Gate Oxide is still present under the Polysilicon whereas the exposed part was etched away, we can conclude that the etching process worked well.

The final structure can be seen by looking at Fig 3.3a.




(a) (b)

Figure 3.3: Structure after the Gate creation (a), enlarged version to highlight the top part of the wafer (b)

## 3.3 LDD

After the Gate creation it was possible to oxidize all the top surface and proceed to the LDD implantation.

As just mentioned an Oxide growth was performed all over the top surface of the wafer. This was necessary as we didn't want that, during the spacers creation, the Gate structure undergoes any possible damage. This process was performed with pure Oxygen for 10 minutes at 900°C.

Once the Gate was protected it was possible to perform the LDD implant. LDD stands for "Lightly Doped Drain" and it's necessary to minimize all the problems created by the so called "hot electrons[4]". To mitigate this phenomenon the region under the Gate and near the Drain was doped with a lower dopant. This decreases the electric field only in this critical interface. The atom used during the implantation was Boron. In Fig. 3.4[5] the projection range is shown as a function of the energy of implantation for Silicon and $SiO_2$. From here it is expected to obtain a 30nm depth for 10keV, which corresponds a peak of distribution in the middle of our Silicon top film. For LDD we decided to start from half this value, which corresponds to 5keV, in order to keep the peak relatively high on surface (we expected it to be around 15 nm, considering a linear approximation). Regarding the dose, we decided to start around $10^{14}$ $cm^{-2}$ based on the planar design.
Eventually, we obtained the best results for an energy of 4keV and a dose of $4 \cdot 10^{14}$ $cm^{-2}$, which are really close to what we calculated.

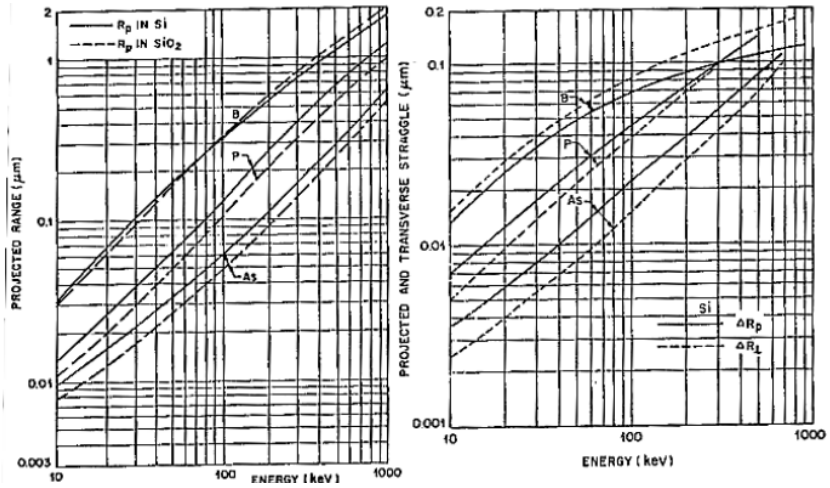



Figure 3.4: Projected range of Si and SiO2 for B, P and As at various energies.

[4]hot electrons are "created" due to the high electric field near the Drain region. This provides enough energy for them to surpass the gate oxide potential barrier causing trapped charge inside the dielectric, thus causing a variation of $V_{th}$ and in physical damage on the oxide. In some exceptions this phenomenon is used to our advantage (FAMOS programming, flash cells), but in general it must be prevented in order to obtain a reliable and usable device

[5]Figure taken from physics of technological process course, prof. Matteo Cocuzza.
Source: *Wolf and Tauber: Silicon Processing for the VLSI Era, Vol. 1, 1986*

After the implantation, an RTA was used to activate the dopant. The RTA lasted 0.1 seconds at 950 °C. Those parameters have been obtained starting from the planar design and tuning them during simulations due to the high complexity regarding the prediction of the resulting doping profiles. In fact, even if all data are known and Fig. 3.5 can be used for the analysis, the $erfc$ function does not allow an easy computation of the depth ($x$ variable).

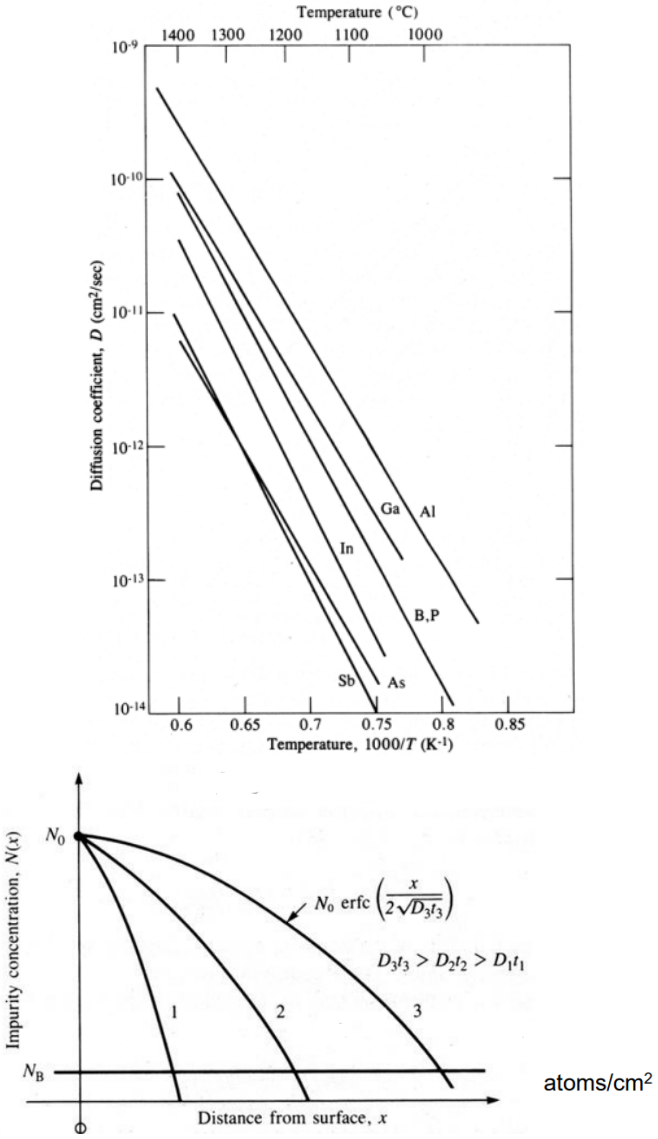
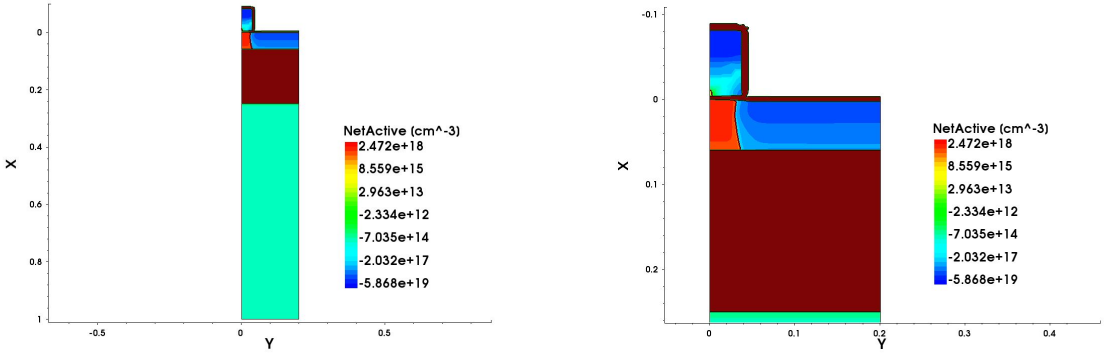



Figure 3.5: Plots to analyze the diffusion of B

The final structure can be seen in Fig. 3.6b.



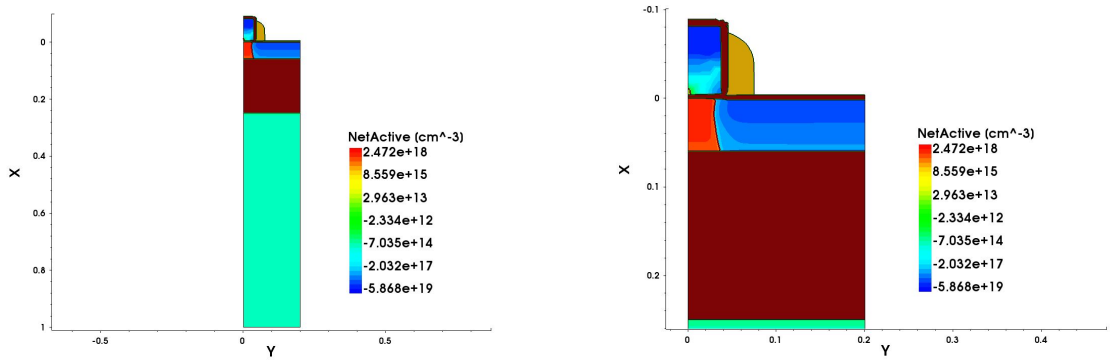

(a) (b)

Figure 3.6: Structure after the LDD process (a), enlarged version to highlight the top part of the wafer (b)

## 3.4 Spacers

Spacers are used to protect the LDD region during the HDD implant. They are made of Silicon Nitride ($Si_3N_4$) because of its different behavior to the same etchants with respect to $SiO_2$, allowing for a good shaping without removing unwanted material.

In fact, in order to create the spacers, one must start depositing an uniform layer of $Si_3N_4$ all over the top surface. Then, with a proper photolithographic process and an anisotropic etching, the unwanted part can be etched away while the required part is kept. The anisotropic etching ensures that the vertical surfaces were etched way slower than the horizontal ones. For this reason, the spacers have the characteristic shape that can be appreciated in Fig. 3.7b.




(a) (b)

Figure 3.7: Structure after the Spacers creation (a), enlarged version to highlight the top part of the wafer (b)

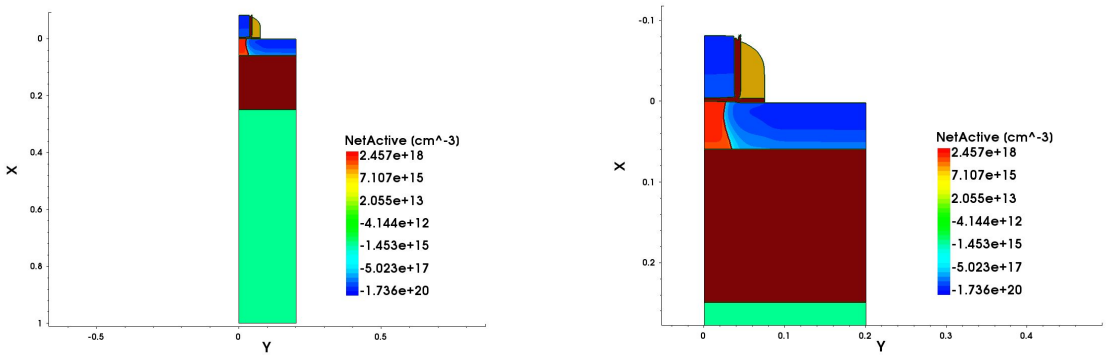
## 3.5 HDD

This step is required to reduce the resistivity of the region and also ensure a good ohmic contact with the Aluminum electrodes that will be putted on top of these region at the end of the fabrication.

Firstly, the Oxide that covered the Gate and the remaining part of the surface was removed (since the Spacers have been already etched there is no more the requirement of a protective layer). The Spacers were actually used as an hard mask to maintain intact the thick layer previously deposited beneath them. The result of this etching is visible in Fig 3.8b.

After the etching the implant was performed. During this process $BF_2$ was used as dopant as Fluorine reduces the Boron mobility during the annealing[3]. Comparing to the LDD step, the energy used was higher in order to increase the projection range (10 keV, around 30nm as said before), the dose was increased to achieve a p+ section ($4.5 \cdot 10^{15}$ cm$^{-2}$) and the process was performed with the wafer tilted by 7°. This tilt was required to minimize the channeling effect as suggested in [4]. To activate and diffuse the dopant a RTA process was performed at the same temperature used for LDD (950 °C) but for a much longer time (10 seconds, 100x).

The final result can also be seen in Fig. 3.8b as well. The difference between the LDD region and the HDD region is evident. The dopant concentration in LDD region goes as low as $10^{15}$ cm$^{-2}$ whereas, in the HDD region, the concentration is $10^{20}$ cm$^{-2}$.




(a) (b)

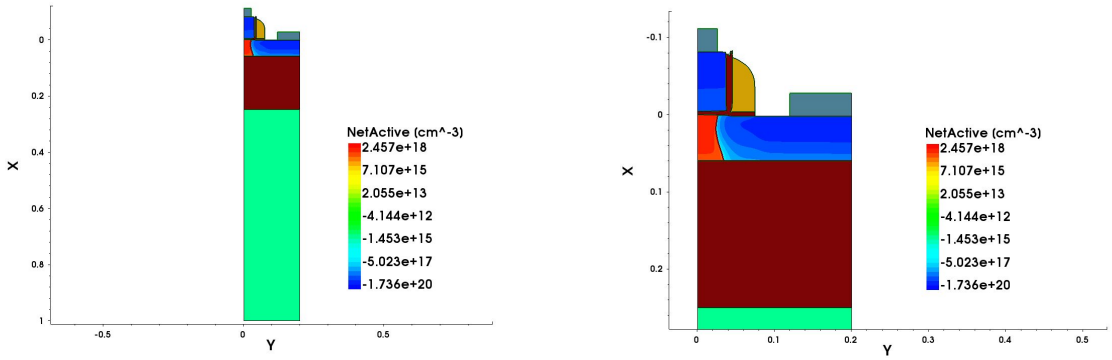
Figure 3.8: Structure after the HDD implant (a), enlarged version to highlight the top part of the wafer (b)

## 3.6 Aluminum contact

In a real case scenario Aluminum electrodes are needed to connect the P-MOS to the various metal layers. In this case no metal layers were simulated or taken into account, the Aluminum contacts were needed only to specify were the simulator had to apply the voltage during the simulation.

This process required the usage of two different photoresists. The positive one was used to cover the Source/Drain contact while the other part of the Aluminum layer, previously deposited, was etched away. The negative one served as a protection while the Gate contact was deposited.

The final result can be seen in Fig 3.9b.

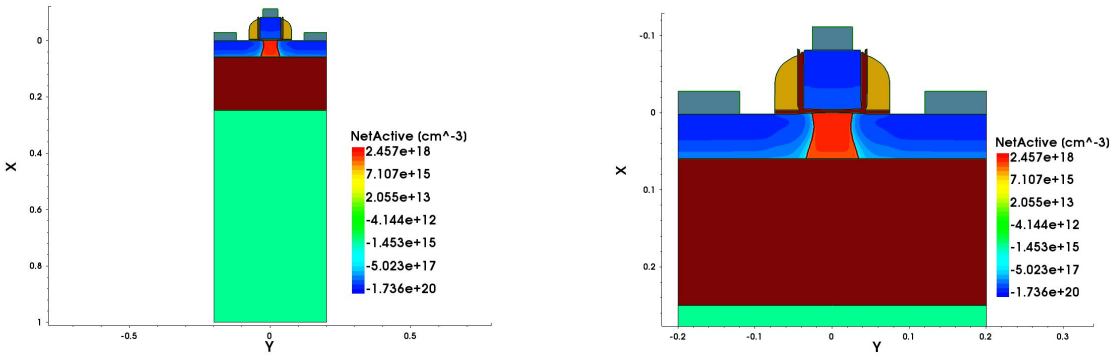



(a) (b)

Figure 3.9: Structure after the Aluminum contact creation (a), enlarged version to highlight the top part of the wafer (b)

## 3.7 Complete structure

At this point the fabrication of the P-MOS is completed. The only thing left to do is the mirroring of the structure along the vertical axis. The final structure in Fig 3.10b shows we obtained a Fully Depleted SOI pMOS transistor.




(a) (b)

Figure 3.10: Complete structure (a), enlarged version to highlight the top part of the wafer (b)

However, when we proceeded with the electrical analysis of the device, we obtained the following results:

| Parameter | Value | Unit of Measurement |
|---|---|---|
| $I_{ON}$ | $8.033 \cdot 10^{-5}$ | A |
| $I_{OFF}$ | $2.817 \cdot 10^{-10}$ | A |
| $I_{ON}/I_{OFF}$ | $2.8 \cdot 10^{5}$ | - |
| $V_{th}\|_{V_{DS}=1V}$ | -0.64 | V |
| $V_{th}\|_{V_{DS}=0.1V}$ | -0.74 | V |
| SS | 106 | mV/dec |
| DIBL | 111 | mV/V |

Table 3.1

It is immediately noticeable how the DIBL is way too far off what we expected and wanted. A good SOI device should have a DIBL close to the ideal value of 60 mV/V, while our transistor has almost twice the ideal value.
Here, we had two possibilities:

- A possible approach is to play with the doping doses in order to obtain a better transition between the Source/Drain regions and the channel. This could have been done by modifying the LDD and HDD processes or by adding an HALO step to the design. Usually for SOI devices this last idea should not be necessary [5], but in this situation it could have been helpful. One last idea could be to increase the doses of Arsenic during the first implantation, since it could be possible that we did not actually achieved a FD-SOI but a DD-SOI [5].

- The second possibility is to tune the buried oxide and top silicon film dimensions, moving towards an UTTB FD-SOI Transistor, and maintain unchanged the doses.

We decided to proceed in the second direction, which means we scaled the silicon top film dimension to $\approx$ 25 nm and the buried oxide to $\approx$ 50 nm. All other processes are maintained unchanged.
In Fig.3.11 it is shown the final result.

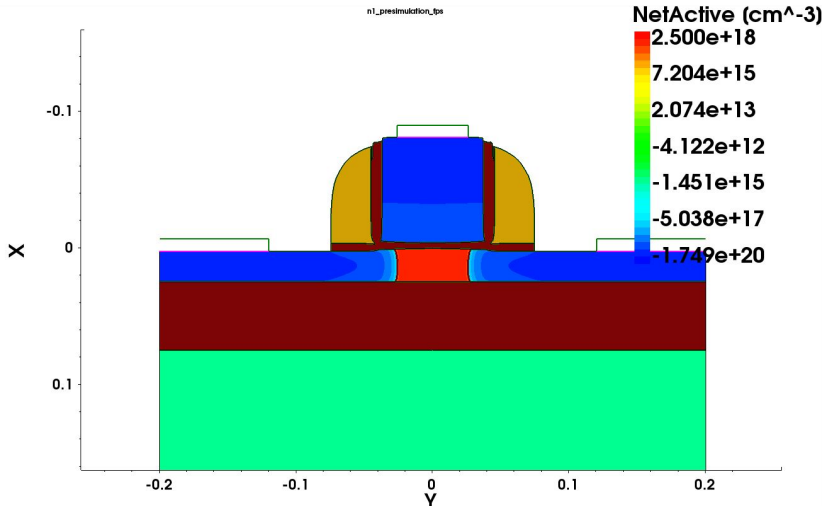



Figure 3.11: Complete structure, UTTB technology

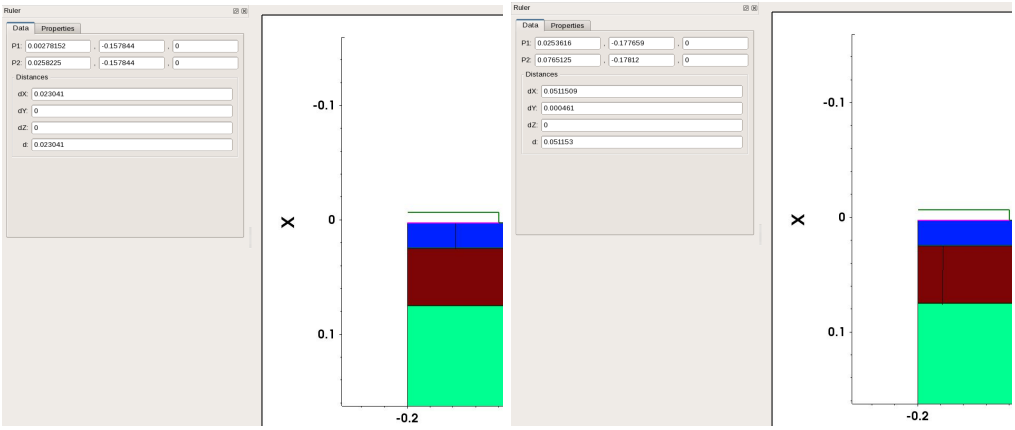



Figure 3.12: Dimensioning of top Si and BOX

# CHAPTER 4

# Electrical Characterisation

In this chapter will be discussed the simulation and the obtained results. Moreover, some considerations about the performance of the P-MOS will be also given.

## 4.1 Simulation

After the fabrication of the structure an electrical simulation was performed. Fig. 4.1 shows the result of the simulation. The IV characteristic plotted is typical of a working P-MOS since at $V_G = 0$ V the transistor is turned off whereas at $V_G$ = -1 V the transistor is turned on.

Once the simulation has ended, all the important FOM (Figure Of Merit) were calculated. All the values that were taken are reported in Tab 4.2.

During the calculation of the most important FOMs we encountered some problems especially when we applied the formulas for SS and $V_{th}$ (we used the same formulas given to us during LAB1). These formulas returned unacceptable values such as $V_{th}$ = -17 mV and SS = 8 $\mu$V/dec. To address this problem we did calculate the SS and the $V_{th}$ graphically, for this reason some errors may have been made. Consequently, when we calculated the value of DIBL (which directly depends on $V_{th}$), the uncertainty in the value of $V_{th}$ was transferred to the DIBL value, making it less precise.

We are not sure about the cause of the incorrect output of the formulas. Different simulations were made to try to fix the problem but it persisted. For this reason we used the just mentioned method to calculate those two values and we compared our plots with research published by others [6, 5] to verify the reliability of our analysis.

Another important FOM is the DIBL[1] (Drain Induced Barrier Lowering). The first simulation was the first made ($V_{DS}$ = 1 V, $V_{GS}$ = -1 V), the second one was made with $V_{DS}$ = 0.1 V, $V_{GS}$ = -1 V. This was necessary in order to acquire another value of $V_{th}$. Here in Fig. 4.1 are shown the two simulations IV characteristics, both on logarithmic scale and on linear scale.

[1] Since the DIBL's value directly depends on $V_{th}$ the uncertainty in the value of $V_{th}$ was transferred to the DIBL value making it less precise

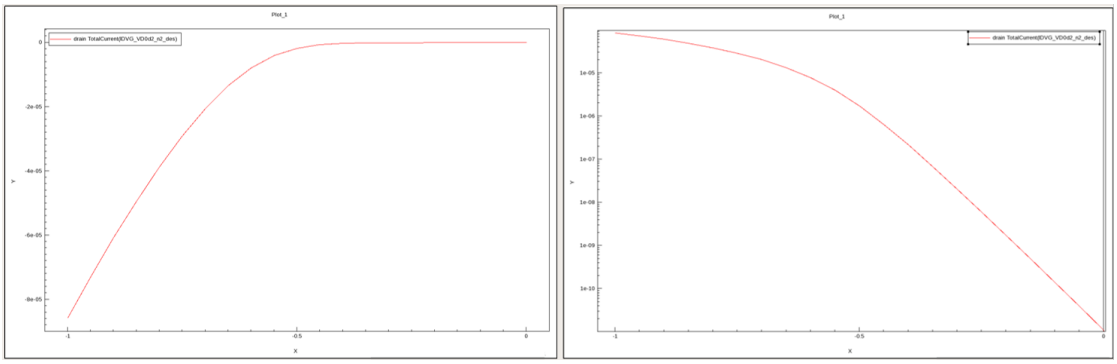


Figure 4.1: I/V characteristic: linear and log scales

## 4.2 FOM's value estimation

In this paragraph will be discussed how we did estimate all the FOMs. For sake of readability at the end a table with all the values will be given.

### 4.2.1 $I_{ON}$ & $I_{OFF}$

The estimation $I_{ON}$ and $I_{OFF}$ was not difficult since the two FOM are defined as:

$$I_{ON} = I_D|_{V_{DS}=VDD, V_{GS}=0V}$$

$$I_{OFF} = I_D|_{V_{DS}=VDD, V_{GS}=VDD}$$

By looking at the graph and by using the tool available inside Sentaurus the two values were taken:

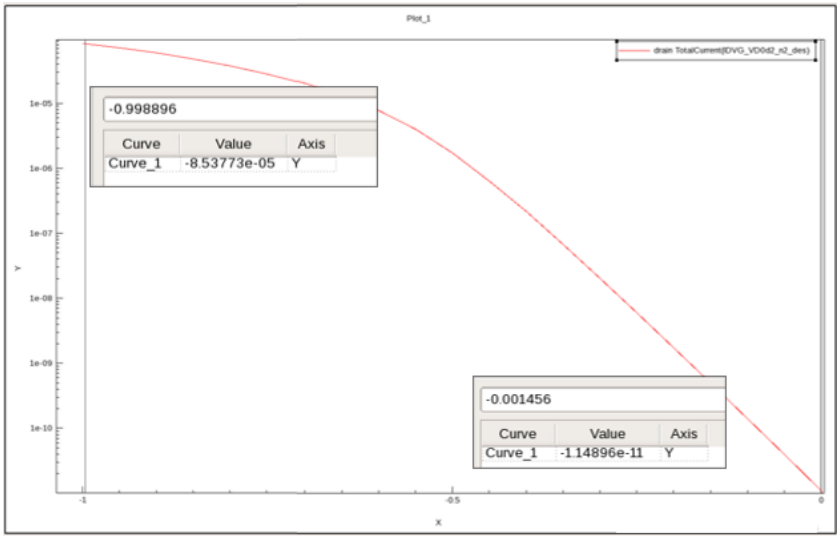



Figure 4.2: $I_{ON}$ & $I_{OFF}$

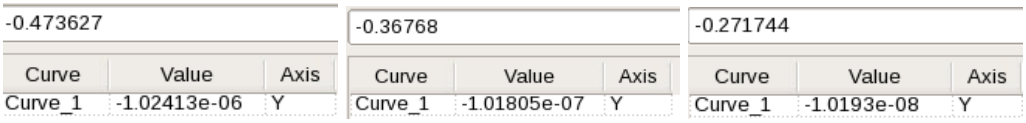
$$I_{ON} = 8.54 \cdot 10^{-5}\ A$$
$$I_{OFF} = 1.15 \cdot 10^{-11}\ A$$

Consequently, their ratio is:

$$\frac{I_{ON}}{I_{OFF}} = \frac{8.54 \cdot 10^{-5}\ A}{1.15 \cdot 10^{-11}\ A} \approx 7.4 \cdot 10^{6}$$

### 4.2.2 SS

As previously discussed we weren't able to obtain the SS value through the given formula. For this reason three different value of $V_{GS}$ were taken at different current values. These values where taken in the simulation with $V_{GS}$ = -1 V and $V_{DS}$ = 1 V. These value are:




| $\mathbf{I_D}$ **(A)** | $\mathbf{V_{GS}}$ **(V)** |
|---|---|
| $1 \cdot 10^{-6}$ | -0.473627 |
| $1 \cdot 10^{-7}$ | -0.36768 |
| $1 \cdot 10^{-8}$ | -0.271744 |

Table 4.1

Since the current values are separated by a decade it's easy to compute the SS value for both decades:

$$SS = \frac{473.627\ mV - 367.680\ mV}{dec} \approx 106\ \frac{mV}{dec}$$
$$SS = \frac{367.680\ mV - 271.744\ mV}{dec} \approx 96\ \frac{mV}{dec}$$

If we take the average the final SS value is:

$$SS = 101\ \frac{mV}{dec}$$

### 4.2.3 $\mathbf{V_{th}}$

The threshold voltage was estimated by looking at the I/V characteristic in the linear version. We examined the region where the P-MOS turns on (when the current starts decreasing significantly as the voltage moves towards lower values) and estimated the $V_{th}$ value. Due to the graphical estimations, some error must be considered to to ensure the reliability of the measurements. We considered an error interval of ±10 mV obtained after various measurements around the same point of analysis.

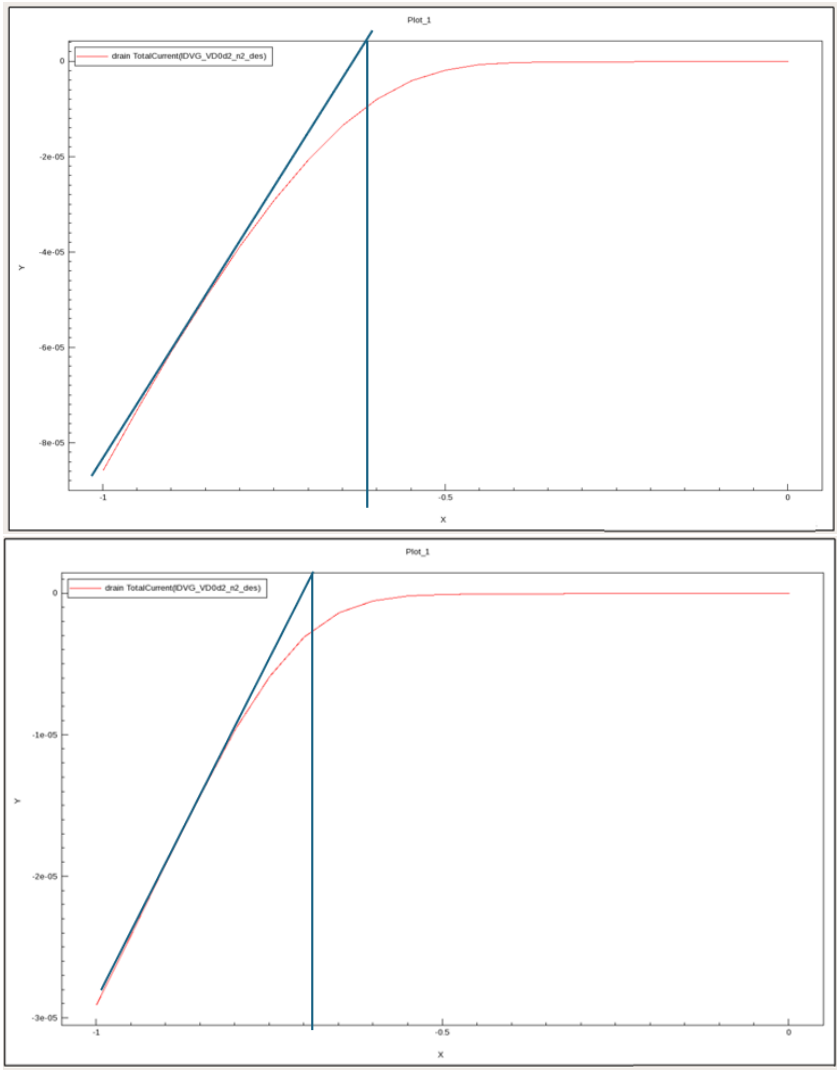



Figure 4.3: graphical measurements of $V_{th}$ for both simulations

As already mentioned, the two simulations are held at ($V_{DS}$ = -1V, $V_{GS}$ = -1V) and ($V_{DS}$ = -0.1V, $V_{GS}$ = -1V). The results we obtained are

$$V_{th}|_{V_{DS}=1V} = (-620 \pm 10)\ mV$$
$$V_{th}|_{V_{DS}=0.1V} = (-680 \pm 10)\ mV$$

The threshold voltage results a higher with what we expected, however it got slightly better when compared to the SOI analysis presented in Table 4.2 and our goal regarding the DIBL was achieved. Therefore, due to the short time remaining until the submission date, we decided to not improve it further. However, our idea to optimize this parameter would have been to try to achieve a better doping transition between channel and Source / Drain regions modifying the doses of the LDD and HDD.

### 4.2.4 DIBL

Drain-Induced Barrier Lowering (DIBL) is a short-channel effect (SCE) through which the threshold voltage of the transistor decreases as the Drain voltage increases. The formula that allows us to calculate the DIBL value is the following one:

$$DIBL = -\frac{V_{th}|_{V_{DS}=1V} - V_{th}|_{V_{DS}=0.1V}}{V_{Drain} - V_{Drain}^{low}} \approx 66.7\ mV/V$$

$$\delta_{DIBL} = \sqrt{\left(\frac{\partial DIBL}{\partial V_{th}|_{V_{DS}=1V}} \cdot \delta_{V_{th}|_{V_{DS}=1V}}\right)^2 + \left(\frac{\partial DIBL}{\partial V_{th}|_{V_{DS}=1V}} \cdot \delta_{V_{th}|_{V_{DS}=1V}}\right)^2} \approx 15.7\ mV$$

$$DIBL = (66.7 \pm 15.7)\ mV/V$$

This result could be optimized in the same way we already mentioned in the $V_{th}$ analysis. However, it was already much better if compared with what we achieved with the standard FD-SOI and since we got short on time we decided to leave it as it is.

### 4.2.5 Values

| Parameter | Value | Unit of Measurement |
|---|---|---|
| $I_{ON}$ | $8.54 \cdot 10^{-5}$ | A |
| $I_{OFF}$ | $1.15 \cdot 10^{-11}$ | A |
| $I_{ON}/I_{OFF}$ | $7.4 \cdot 10^{6}$ | - |
| $V_{th}\|_{V_{DS}=1V}$ | -0.62 | V |
| $V_{th}\|_{V_{DS}=0.1V}$ | -0.68 | V |
| SS | 101 | mV/dec |
| DIBL | 66.7 | mV/V |

Table 4.2

## 4.3 Considerations on the obtained values

To have a good reference, as already mentioned, we compared the obtained value with [5, 6].

On a first glance, the plots look very close to what we achieved. $I_{ON}$ is in the same order of magnitude in respect to the papers show wheres $I_{OFF}$ is a bit worse if compared to what they declared. However, we did not fully understand how [6] obtained the declared value by looking at their plot and the ratio $I_{ON}/I_{OFF}$ is what really matters and shows a more than acceptable device. We therefore concluded our results were acceptable.

Both $V_{th}|_{V_{DS}=1V}$ and $V_{th}|_{V_{DS}=0.1V}$ are larger than what was wanted (see Fig 2.1) and what the paper [6] declare, but again their plot shows a completely different result, much closer to what we got, so we assumed acceptable this point of analysis too. Same can be said regarding the DIBL.

In any case, these issues and how we would have proceed in order to minimize them have already been discussed before.

SS has a larger value than the example presented in the paper [6] but it very close to it. For this reason this value was considered acceptable and no further optimization were made to decrease it.

# CHAPTER 5

# Conclusion

The SOI P-MOS was fabricated with the requested channel length and then its behaviour was simulated. All the results obtained are similar to what was predicted during the theoretical calculations or with what was reported in the articles cited at the end of last chapter.

# APPENDIX A

# Planar pMOS

Before working on the SOI pMOS device, we started by designing a planar pMOS transistor in order to have an idea of what could have been reasonable parameters to set on the various processes.
In the first laboratory of this course, we worked with a n-type planar transistor, so our idea was just to build the corrispective p-type tuning properly the various doping doses and energies and the annealing threatments.
It is well known knowledge that the various dopants do not have the same physical characteristics, therefore atoms behave differently to diffusion and implantation processes: to pass from an n-type transistor to a p-type maintaining similar performances is not just a manner of substituting every Boron with Arsenic or Phosphorus and viceversa, but it is required to re-tune most of the parameters.
Here it is briefly listed all the step we did to create the planar transistor and are shown the figures of the final structure and of the simulations plot. Regarding these last ones, the number of points used has been minimized to 10, since it was not necessary to analyze them in details but to just have an idea if the device was properly working or not.

| Step | Description |
|---|---|
| **Substrate definition** | Boron $1.5 \cdot 10^{15}$ $cm^{-3}$ |
| **Substrate implantation 1** | Arsenic $1.0 \cdot 10^{12}$ $cm^{-2}$, 200 keV |
| **Substrate implantation 2** | Arsenic $1.0 \cdot 10^{12}$ $cm^{-2}$, 150 keV |
| **Substrate implantation 3** | Arsenic $1.0 \cdot 10^{12}$ $cm^{-2}$, 100 keV |
| **Substrate implantation 4** | Arsenic $1.0 \cdot 10^{12}$ $cm^{-2}$, 70 keV |
| **Substrate implantation 5** | Arsenic $1.0 \cdot 10^{12}$ $cm^{-2}$, 50 keV |
| **Substrate implantation 6** | Arsenic $2.0 \cdot 10^{12}$ $cm^{-2}$, 20 keV |
| **RTA1** | Temperature 1000°C, 60 s |
| **Dry oxidation** | Temperature 850°C, 10 min |
| **PolySilicon deposition and etching** | |
| **Dry oxidation** | Temperature 900°C, 10 min |
| **LDD** | Boron $1 \cdot 10^{13} cm^{-2}$, 5 keV |
| **HALO** | 4x Arsenic $0.25 \cdot 10^{12} cm^{-2}$, 20 keV, tilt = 30°, rotation = 90° |
| **RTA2** | Temprature 950°C, 5 s |
| **Nitride deposition and etching** | |
| **HDD** | Boron $4.5 \cdot 10^{15} cm^{-2}$, 15 keV |
| **Aluminum deposition and etching** | |

Table A.1: Planar pMOS steps

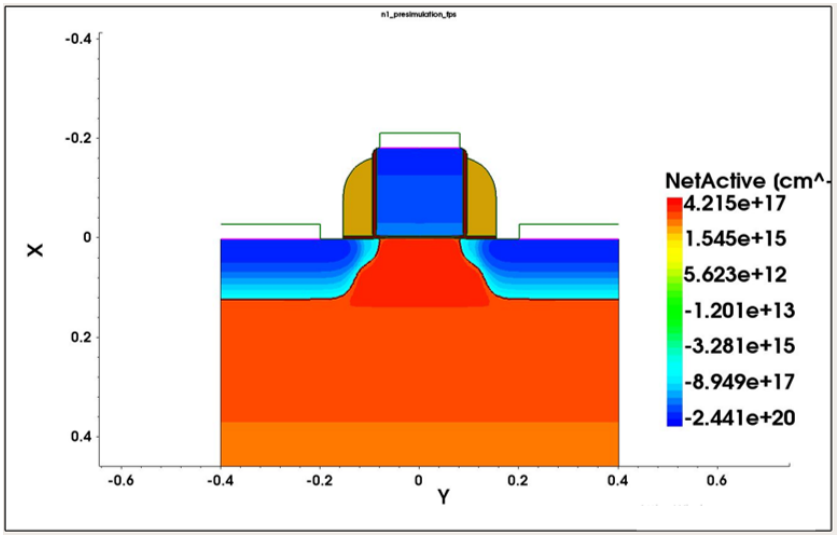



Figure A.1: Structure of the pMOS device

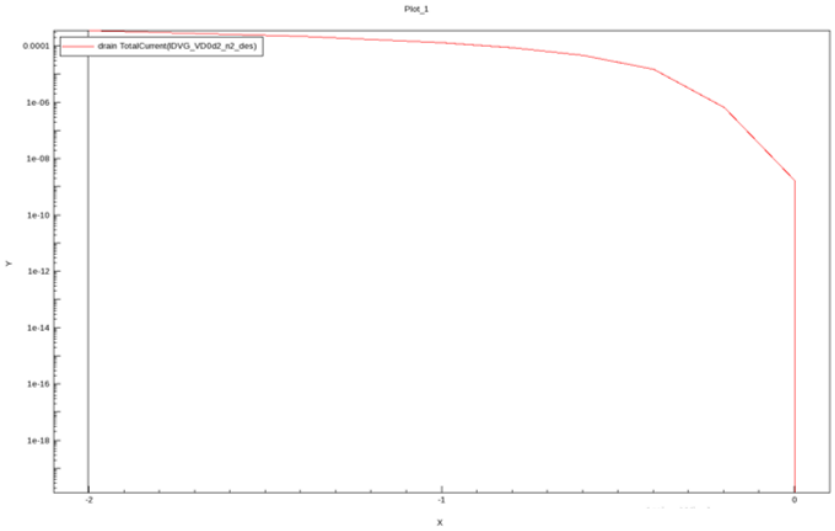



Figure A.2: Logarithm I/V characteristics

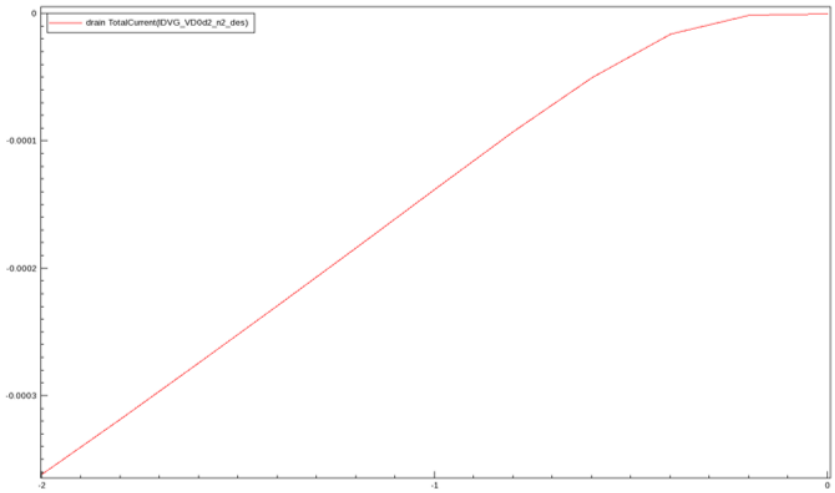



Figure A.3: Linear I/V characteristics

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